4MA1_1H_ch3_sequences_functions_graphs

Nguyen Khac Kiem PhD

1. 4MA1/1H_Winter_2024_Q25

The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geqslant 6$

Find the inverse function $f^{-1}(x)$

4 marks

2. 4MA1/1H_Winter_2024_Q23

The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)
The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k
Show clear algebraic working.

5 marks

3. 4MA1/1H_Winter_2024_Q19

An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

5 marks

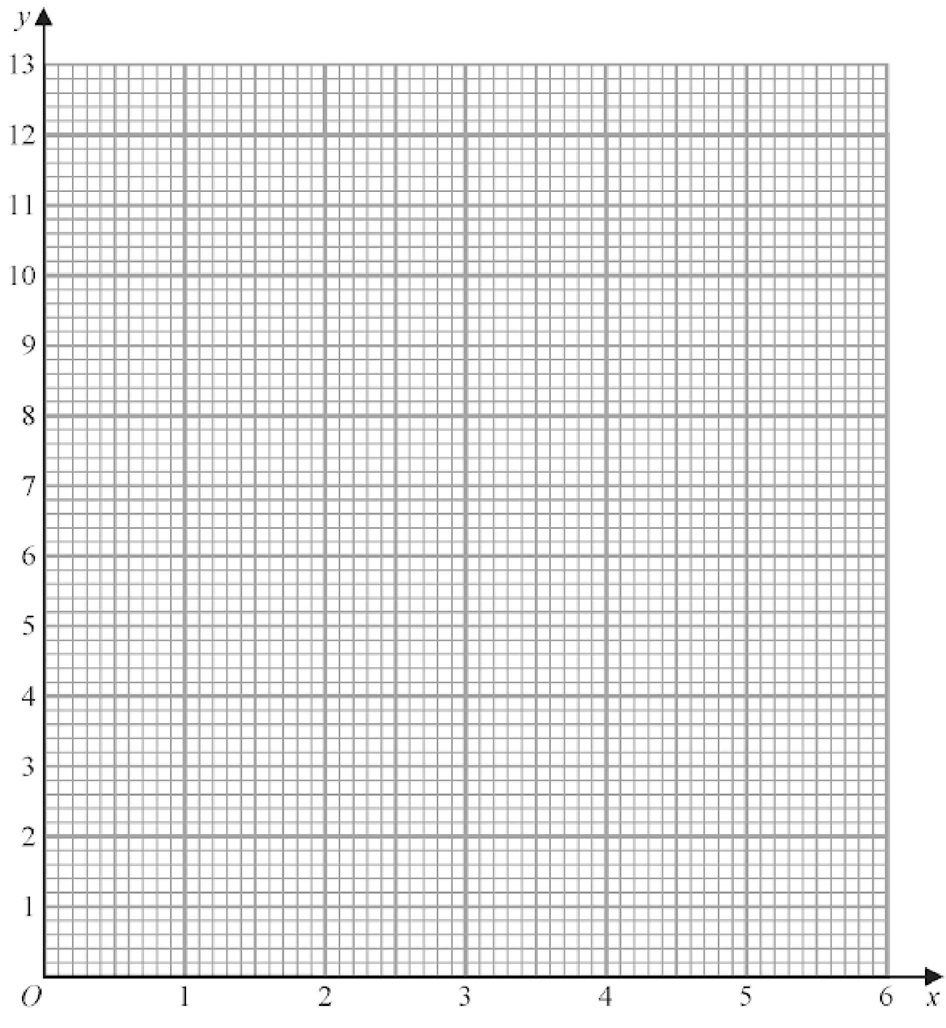
4. 4MA1/1H_Winter_2024_Q11

(a) Complete the table of values for $y = 2\left(x + \frac{1}{x}\right)$

x	0.5	1	2	3	4	5	6
y			5	6.7			12.3

(2)

(b) On the grid, draw the graph of $y = 2\left(x + \frac{1}{x}\right)$ for $0.5 \leqslant x \leqslant 6$



(2)

5. 4MA1/1HR_Summer_2024_Q23

Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.
Show clear algebraic working.

(4)

6. 4MA1/1HR_Summer_2024_Q20

(a) Express $2x^2 - 11x + 9$ in the form $a(x-b)^2 - c$ where a, b and c are numbers to be found.

(3)

The curve **C** has equation $y = 2(x-3)^2 - 11(x-3) + 9$

The point P is the minimum point on **C**

(b) Find the coordinates of P

(2)

7. 4MA1/1HR_Summer_2024_Q19

A curve C has equation $y = x^3 - 8x^2 - 12x + 5$

Curve C has exactly two stationary points, one at point A and one at point B such that

x coordinate of point A > x coordinate of point B

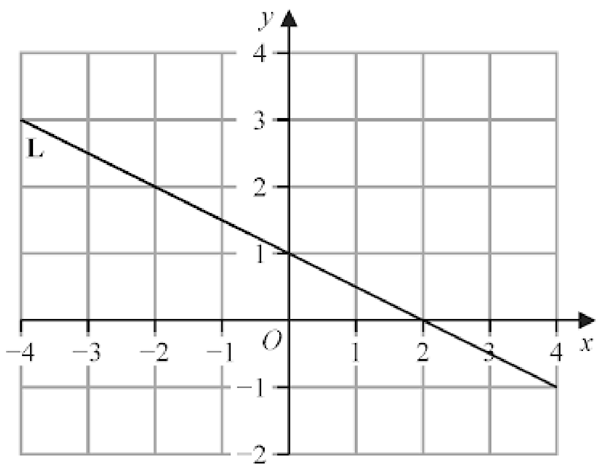
Find the coordinates of point A
Show clear algebraic working.

(.................... , ...)

(5)

8. 4MA1/1HR_Summer_2024_Q9

Line **L** is drawn on the grid.



Find an equation for **L**
Give your answer in the form $y = mx + c$

(3)

9. 4MA1/1H_Summer_2024_Q24

A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°
The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.
Show clear algebraic working.

.... °

(6)

10. 4MA1/1H_Summer_2024_Q15

The function f is defined as

$$f : x \mapsto \frac{3x+1}{x-2}$$

(a) State the value of x that cannot be included in any domain of the function f

(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(3)

11. 4MA1/1H_Summer_2024_Q1

Here are the first four terms of an arithmetic sequence.

$$1 \qquad 4 \qquad 7 \qquad 10$$

(a) Find an expression, in terms of n, for the nth term of this sequence.

(2)

The nth term of a different arithmetic sequence is $5n + 17$

(b) Find the 12th term of this sequence.

(1)

12. 4MA1/1HR_Winter_2023_Q24

An arithmetic sequence has first term 8 and common difference 11
The sequence has k terms, where $k > 21$

The sum of the last 20 terms of the sequence is 10 170

Find the value of k
Show clear algebraic working.

$k =$ ....

5 marks

13. 4MA1/1HR_Winter_2023_Q16

The function f is such that

$$f(x) = \frac{2}{3x - 5} \qquad \text{where } x \neq \frac{5}{3}$$

(a) Find $f\left(\frac{1}{3}\right)$

....

(1)

(b) Find $f^{-1}(x)$

$f^{-1}(x) =$ ....

(2)

The function g is such that

$$g(x) = 5x^2 - 20x + 23$$

(c) Express $g(x)$ in the form $a(x - b)^2 + c$

....

(3)

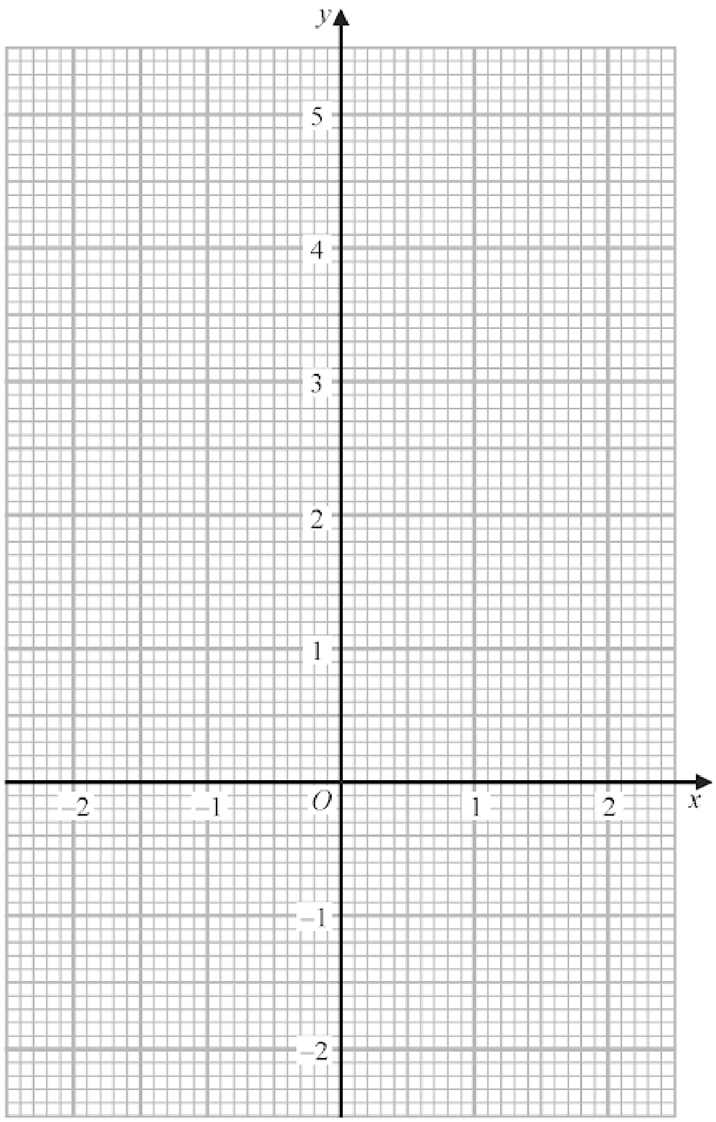
14. 4MA1/1HR_Winter_2023_Q15

(a) Complete the table of values for $y = x^3 - 3x + 2$

x	-2	-1	-0.5	0	1	1.5	2
y		4	3.4		0	0.9	

(2)

(b) On the grid, draw the graph of $y = x^3 - 3x + 2$ for values of x from -2 to 2



(2)

(c) By drawing a suitable straight line on the grid, use your graph to find an estimate for the solution of

$$2x^3 - 3x + 4 = 0$$

Give your answer correct to one decimal place.

....
(3)

15. 4MA1/1H_Winter_2023_Q23

$ABCD$ is a kite.

$AB = AD$ and $CB = CD$

The point B has coordinates $(k, 1)$ where k is a negative constant.
The point D has coordinates $(8, 7)$

The straight line **L** passes through the points B and D

The straight line **L** is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC
Give your answer in the form $px + qy = r$ where p, q and r are integers.
Show your working clearly.

16. 4MA1/1H_Winter_2023_Q22

The first term of an arithmetic series is $(2t + 1)$ where $t > 0$
The nth term of this arithmetic series is $(14t - 5)$

The common difference of the series is 3

The sum of the first n terms of the series can be written as $p(qt - 1)^r$ where p, q and r are integers.

Find the value of p, the value of q and the value of r
Show clear algebraic working.

17. 4MA1/1H_Winter_2023_Q18

The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) = \ldots\ldots\ldots\ldots\ldots\ldots$

(1)

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2 - x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

$\ldots\ldots\ldots\ldots\ldots\ldots$

(1)

(ii) Express the composite function hg in the form $hg(x) = \ldots$
Give your answer in its simplest form.

$hg(x) = \ldots\ldots\ldots\ldots\ldots\ldots$

(2)

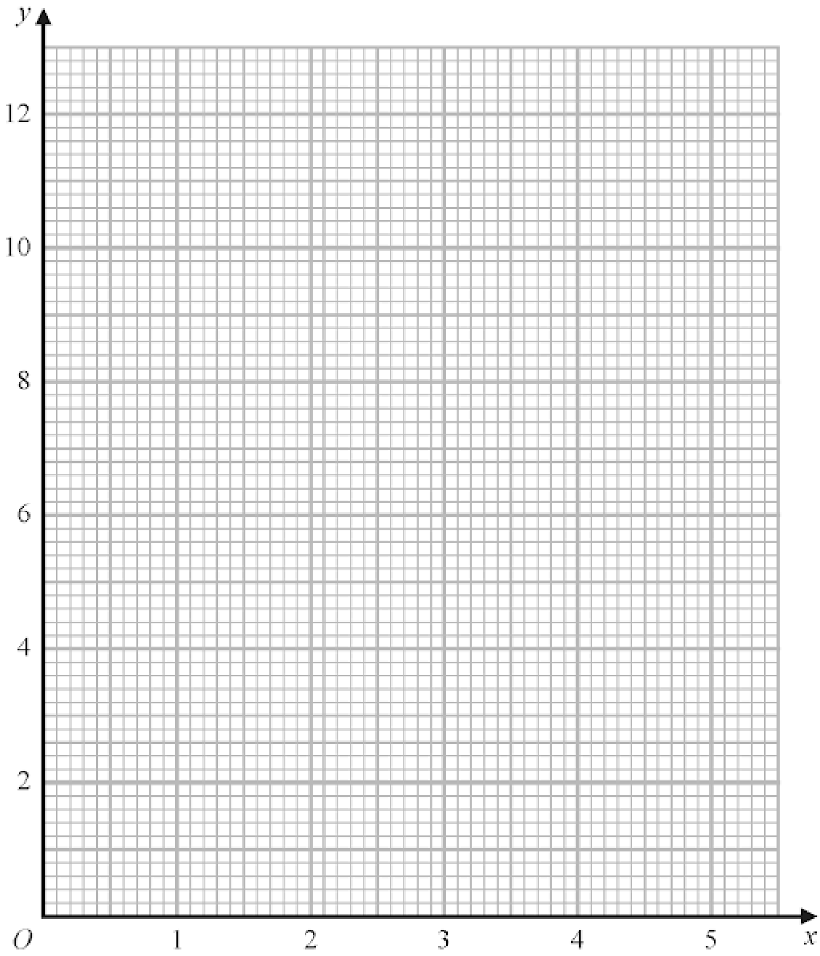
18. 4MA1/1H_Winter_2023_Q14

(a) Complete the table of values for $y = \frac{2}{x}\left(5 - \frac{1}{x}\right)$

x	0.5	1	2	3	4	5
y		8		3.1	2.4	1.9

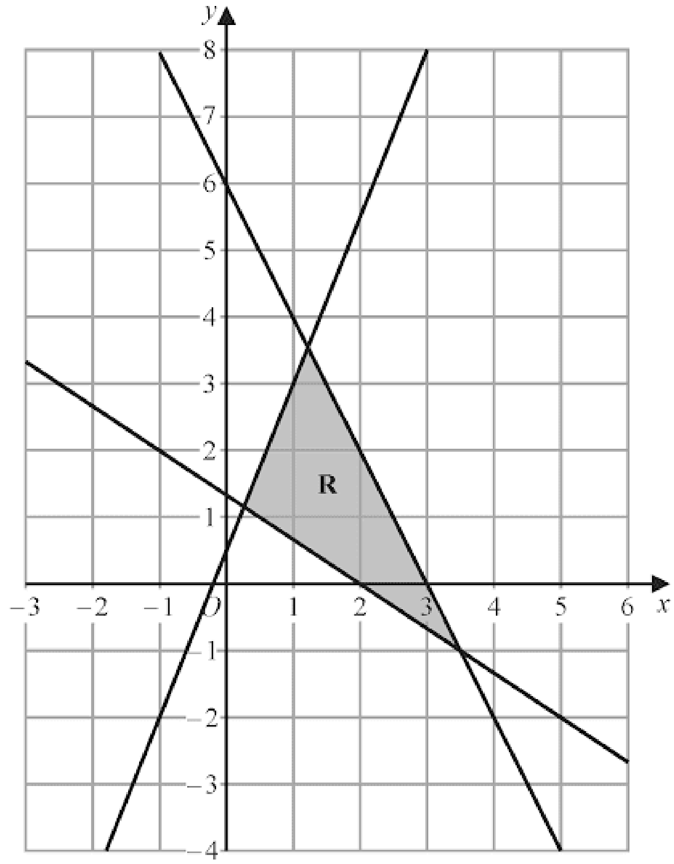
(1)

(b) On the grid, draw the graph of $y = \frac{2}{x}\left(5 - \frac{1}{x}\right)$ for $0.5 \leqslant x \leqslant 5$



(2)

19. 4MA1/1H_Winter_2023_Q11



The region **R**, shown shaded in the diagram, is bounded by the straight lines with equations

$2x + y = 6$

$2y = 5x + 1$

$3y + 2x = 4$

Write down the three inequalities that define **R**

....

....

....

3 marks

Curve **C** has equation $y = f(x)$
The graph of curve **C** has one maximum point.
The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.................... , ...)
(1)

(ii) $y = f(x) - 7$

(.................... , ...)
(1)

(iii) $y = f(-x)$

(.................... , ...)
(1)

Curve **L** has equation $y = x^2 + 7x + 20$

Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

(b) Find an equation for **S**
Give your answer in the form $y = ax^2 + bx + c$

$y =$ ....
(4)

21. 4MA1/1HR_Summer_2023_Q21

A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.................... , ...)
(1)

(b) $y = f(3x)$

(.................... , ...)
(1)

22. 4MA1/1H_Summer_2023_Q20

The sum of the first 80 terms of an arithmetic series, S, is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X
Show your working clearly.

23. 4MA1/1H_Summer_2023_Q17

The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \qquad x \geqslant 0$$

(a) What value of x must be excluded from any domain of g?

....

(1)

(b) Solve $gh(x) = 1$

....

(3)

24. 4MA1/1H_Summer_2023_Q13

A is the point with coordinates (–5, 12)

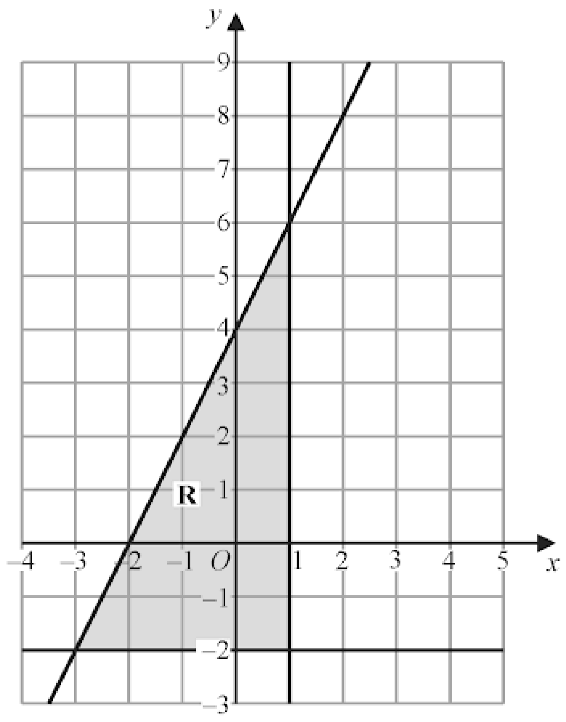
B is the point with coordinates (19, –48)

Find an equation of the straight line that passes through the points A and B

....

3 marks

25. 4MA1/1H_Summer_2023_Q7



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Find the inequalities that define **R**

....

....

....

4 marks